

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

**39EJ DEMINERALIZATION SYSTEM-CIVIL BUILDING & FDN. DESIGN
DESIGN CALCULATION**

ISSUED FOR CONSTRUCTION

MD-216-39EJ-EG-CV-CAL-1459

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer	Ma'aden Document Number:
D01	16-10-2023	AS BUILT	TEC	AVG	SJM		MD-216-39EJ-EG-CV-CAL-1459 D01
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 MA'ADEN شركة التعدين العربية السعودية Saudi Arabian Mining Company		 Outotec		 LARSEN & TOUBRO		DESIGN DOCUMENT Ma'aden Document Number: MD-216-39EJ-EG-CV-CAL-1459 Outotec Document Number: 900707_ GAU01_AG36_CED01_39304_OU100703853 L&T Document Number: O19045_450_C_K_24434_001					PAGE 2 of 31	
Global Project ID		900707										
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Approved		SJM	SJM	SJM			Approved	SJM	SJM	SJM		
REFERENCE DOCUMENTS												
DOCUMENT NUMBER		DOCUMENT DESCRIPTION										
MD-216-1000-EG-ST-CRT-1001		Civil Structural Design Criteria										
SBC 301 :2007		Saudi Structural - Loading & Forces Code										
SBC 306 :2007		Saudi Steel Structures Code										
AISC LRFD		Building Code Requirement of Steel Structural Building										
AISC Steel Design Guide 1 - 2nd Edition		Base Plate and Anchor Rod Design										
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Approved		SJM	SJM	SJM		Approved		SJM	SJM	SJM

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

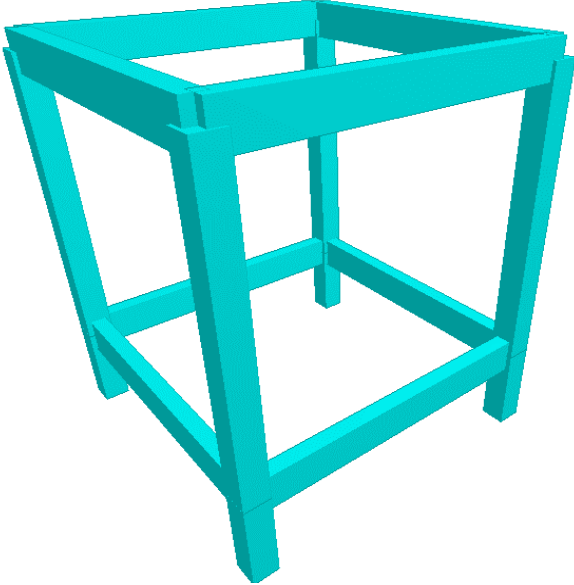
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1. INTENT The document is prepared for design of Super structure and Foundation of a Civil building 2. EXECUTIVE SUMMARY The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project. MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarrah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim. The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week. 3. SCOPE The document is prepared for design of Super structure and Foundation of a Civil building All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007 3D- Model and analysis have been performed in Staad-Pro Software. Foundation has been design using MS-Excel as Per SBC-303:2007, SBC-304:2018 & ACI 318-08. 4 DESIGN OF RCC ELEMENTS 4.1 Design Design of RCC Str. and its components shall be designed in accordance with [1] SBC-301-2007-Structural Loading & Forces. [2] SBC-303-2007-Structural Soil & Foundations. [3] SBC-304-2018-Structural Concrete Structures. [4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete. [5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete [6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures. [7] ACI 318-08 Building Code Requirements. [8] PCI –MNL-120-04 –Design Handbook, 6th edition. 4.2 Materials Compressive Strength of Concrete at 28th days (f_c') = 35 MPa Yield Strength of Steel (f_y) = 420 MPa 4.3 Reference Reference is made to codes, standards and specifications as indicated in Civil, Structural and Dwg. No. MD-216-3310-EG-CV-DED-1459 5 LOAD CALCULATONS 5.1 Dead Load (DL) Selfweight of structure is applied through Staad Pro. Thickness of Roof Slab @ 225mm = 5.625 kN/Sqm (Vertical) Load due to 250 thk and 4 m ht block wall = 20 kN/m (Vertical) 5.2 Live Load (LL) .Live load on accessible Roof @ 150kg/m2 = 1.5 kN/Sqm (Vertical) 5.3 Wind Load (WL) Wind Load Calculation Wind Load Parameters (As per design Criteria) : Basic wind speed, V_b = 43 m/s = 155 Km/h Exposure category = C (For Process Plant Area) Building Category = II (Ref. Table 1.6-1 : SBC 301-2007) Wind directionability factor, K_d = 0.85 (Ref. Table 6.4-1 : SBC 301-2007) Topography Factor, K_{zt} = 1.00 (As The project is not located in isolated hills or escarpments) Importance Factor, I = 1.00 (Ref. Table 6.5-1 : SBC 301-2007) Velocity Pressure Co-efficient = K_z (Ref. Table 7.2-2 : SBC 301-2007) Gust Effect Factor, G = 0.85 (For Rigid Structure) (Ref. Section 7.2.7.1 : SBC 301-2007) Force Co-efficient, C_f = 1.30 Pressure On Wall = 1.08 kN/m² Force on Column = 2.70 kN/m Wind Velocity Pressure, q_z = 0.0473x10⁻³K_zK_{zt}K_dV²I (kN/m²) (Ref. Eq. 7.2-12 : SBC 301-2007) <table><tr><th>Height</th><th>kz</th><th>Pressure</th><th>Unit</th></tr><tr><td>5 m</td><td>0.86</td><td>0.831</td><td>kN/m²</td></tr><tr><td>6 m</td><td>0.90</td><td>0.869</td><td>kN/m²</td></tr><tr><td>8 m</td><td>0.95</td><td>0.918</td><td>kN/m²</td></tr><tr><td>10 m</td><td>1.00</td><td>0.966</td><td>kN/m²</td></tr></table> Pressure on Column, q_z = 0.83 kN/m²																				Height	kz	Pressure	Unit	5 m	0.86	0.831	kN/m ²	6 m	0.90	0.869	kN/m ²	8 m	0.95	0.918	kN/m ²	10 m	1.00	0.966	kN/m ²
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5.4 Teampature Load (TL)		There are two temperature Load case is considered that is TL1 & TL2										
		TL1 - For Temperature Difference + 25 ⁰ C										
		TL2 - For Temperature Difference -25 ⁰ C										
5.5 Earthquake Load (EQL)												
Seismic Design Category		= A										
Site class		= B It is corresponds to Site Class 2 in Staad										
Spectral response acceleration (1-sec)		S1	=	0.02								
Spectral response acceleration (.02-sec)		Ss	=	0.1								
Importance factor		I	=	1 (From Table 9.5, SBC 301:2007)								
Response Reduction factor along X		Rx	=	2.5 (From Table 10.2, SBC 301:2007)								
Response Reduction factor along Z		Rz	=	2.5 (From Table 10.2, SBC 301:2007)								
Height of building		h	=	5 m								
Time period along X		Tx	=	0.18 (eqn. 10.9.3.2-1, SBC 301:2007)								
Time period along Z		Tz	=	0.18 (eqn. 10.9.3.2-1, SBC 301:2007)								
Factor Fa		Fa	=	1.6								
Factor Fv		Fv	=	2.4								
SM1		Fa x S1	=	0.032								
SMS		Fa x Ss	=	0.16								
Design spectral acceleration (1-sec)		SD1 = 2/3 x SM1	=	0.0213								
Design spectral acceleration (0.2-sec)		SDS = 2/3 x SMS	=	0.1067								
		The above parameters are used in the staad input to find the final earthquake load of the structure as per IBC-2003										
5.6 Load Combinations												
5.6.1 Factored Load Combination for Structural design		(As Per Section 2.3 of SBC 301-2007)										
5.6.2 Load Combination for Servicibility check												
1.1.4 DL		1. DL										
2. 1.4 DL + 1.7 LL+ 0.5 LR		2. DL + TL + LL										
3. 1.2 DL+ 1.6 WL + 1.0 LL + 0.5 LR		3. DL + 0.75 LL +0.75 WL										
4. 1.2 DL +1.0 EL + 1.0 LL		4 . DL + 0.7 E+ 0.75 L										
5. 1.2 DL + 1.2 TL+ 1.6 LL		5 .0.6 DL +1.0 WL										
6. 0.9 DL + 1.6 WL		6. 0.6 DL + 0.7 EL										
7. 0.9 DL + 1.0 EL												

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6 GEOMETRY

6.1 Geometry of the civil Building



3D-View

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7 DESIGN OF RCC ELEMENTS										
All Major components i.e. RCC Column, Beam, Wall have been design in Staad-Pro. For Design Results, Refer Attachemnt 3 - Staad Output file. Foundation has been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.										

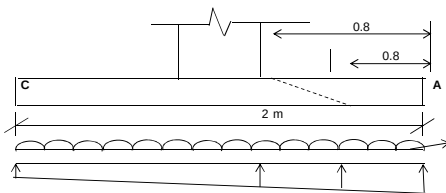
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7.1 DESIGN OF FOUNDATION MKD. F1 : (NODE NO. : 13,14,15 & 16) Net Safe Bearing Capacity of Soil = 20 Mton/m² Depth of Foundation from FGL = 1.5 m Gross Allowable Bearing Capacity of Soil = 20 + 1.5 X 1.9 = 22.85 Mton/m² Compressive Strength of Concrete , fc' = 35 N/mm² Yield Strength of Main Reinf. , fy = 420 N/mm² Yield Strength of Shear Reinf. , fyt = 420 N/mm² Strength Reduction Factor of Main Reinf. , Φ = 0.9 Strength Reduction Factor of Shear Reinf. , Φt = 0.75 Clear Cover = 50 mm Density of Concrete = 2.5 Mton/m³ Density of Dry Soil = 1.9 Mton/m³ Density of Saturated Soil = 2 Mton/m³ Density of Water = 1 Mton/m³ FOUNDATION DIMENSION : Length of Foundation (Lx) = 2 m Width of Foundation (Lz) = 2 m Thickness of Base Raft = 0.4 m Length of Column (lx) = 0.4 m Width of Column (lz) = 0.4 m Area , A = Lx*Lz = 4 m² Zxx = Lx*Lz²/6 = 1.333 m³ Zzz = Lz*Lx²/6 = 1.333 m³																																																																																																																									
WT. OF FOUNDATION (Wf) : <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>ITEM</th> <th>DRY CONDITION (Mton)</th> </tr> <tr> <td>Raft =</td> <td>2X2X0.4X2.5 = 4</td> </tr> <tr> <td>Soil =</td> <td>((2X2)-(0.4X0.4))X1.1X1.9 = 8.026</td> </tr> <tr> <td>Total =</td> <td>12.026</td> </tr> </table>					ITEM	DRY CONDITION (Mton)	Raft =	2X2X0.4X2.5 = 4	Soil =	((2X2)-(0.4X0.4))X1.1X1.9 = 8.026	Total =	12.026																																																																																																													
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OVERTURNING CHECK: L/C : 30 Max Upward Force = 0 Mton Self Wt. of Foundation = 12.026 Mton Overturning Moment = 0 X 2/2 + 5.57142857142857 = 5.571 Mton-m Resisting Moment = 12.026 X 0.9 X 2/2 = 10.823 Mton-m F.O.S = 1.94 > 1.5 O.K.																																																																																																																									

  		DESIGN DOCUMENT Ma'aden Document Number: MD-216-39EJ-EG-CV-CAL-1459 Outotec Document Number: 900707_GAU01_AG36_CED01_39304_OU100703853 L&T Document Number: 019045_450_C_K_24434_001	PAGE 10 of 31
Global Project ID	900707		
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION		
Project Name	MANSOURAH MASSARAH GOLD PROJECT		
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)		
Project Country	KINGDOM OF SAUDI ARABIA		
Ma'aden Revision No:	A01	C01	
Date	20-10-2020	20-11-2020	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	
Prepared By	TEC	TEC	
Checked	AVG	AVG	
Approved	SJM	SJM	
Outotec / L&T Revision No.:	0	1	
Date	20-10-2020	20-11-2020	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	
Prepared By	TEC	TEC	
Checked	AVG	AVG	
Approved	SJM	SJM	

ONE-WAY SHEAR CHECK

$V_u = 1.853 \text{ Mton}$
 $Nominal \text{ Shear Strength}, V_c = \min [(a) , (b) , (c)]$
 $[a] = (0.16 \lambda^* f_c^* 0.5 + 17 P^* V_u^* d_{eff} / (Mu)) * b^* d_{eff}$
 $= (0.16 \times 1 \times 35^* 0.5 + 17 \times 0.0022 \times 1.853 \times 10^4 \times 344 / (6.071 \times 10^7)) \times 1000 \times 344 / 10^4$
 $= 32.697 \text{ Mton}$
 $[b] = (0.16 \lambda^* f_c^* 0.5 + 17 P^*) * b^* d_{eff}$
 $= (0.16 \times 1 \times 35^* 0.5 + 17 \times 0.0022) \times 1000 \times 344 / 10^4$
 $= 33.849 \text{ Mton}$
 $[c] = 0.29 \lambda^* f_c^* 0.5 * b^* d_{eff}$
 $= 0.29 \times 1 \times 35^* 0.5 \times 1000 \times 344 / 10^4$
 $= 59.019 \text{ Mton}$
HENCE, $V_c = 32.697 \text{ Mton}$
 $\Phi^* V_c = 0.75 \times 32.697$
 $= 24.523 \text{ Mton}$
 $\Phi^* (V_c + 0.66 f_c^* 0.5 * b^* d_{eff}) = 0.75 \times (32.697 \times 10^4 + 0.66 \times 35^* 0.5 \times 1000 \times 344) / 10^4$
 $= 103.191 \text{ Mton}$

$\lambda = 1$
 $> 1.853 \text{ Mton}, \text{ Hence O.K.}$
 $> V_u = 1.853 \text{ Mton}, \text{ CROSS SECTION O.K.}$

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :


L/C	Pressure at Point A (Mton/m ²)	Pressure at Point C (Mton/m ²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
19	19.393	9.375	15.39	15.39	6.071	0.728
19	19.393	9.375	15.39	15.39	6.071	0.728
10	12.47	10.398	11.64	11.64	3.179	2.074
17	7.987	7.773	7.9	7.9	1.586	1.471
19	19.393	9.375	15.39	15.39	6.071	0.728
19	19.393	9.375	15.39	15.39	6.071	0.728
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19	19.393	9.375	15.39	15.39	6.071	0.728
13	9.93	9.72	9.85	9.85	2.206	2.094
14	9.93	8.14	9.21	9.21	2.342	1.388
19	19.393	9.375	15.39	15.39	6.071	0.728
19	19.393	9.375	15.39	15.39	6.071	0.728
				MAX =	6.071	2.094

ALONG Z - DIRECTION :
 $M_u = 6.071 \text{ T-m}$
 $\Phi^* K_n = \frac{M_u}{b d^2} = \frac{6.071 \times 10^7}{(1000 \times 344^2)} = 0.513 \text{ N/mm}^2$
 $P_t = 0.0014$
 $\text{Min. } P_t = 0.0018$
 $\text{Provided } 12 \text{ mm @}$
 $\text{Reqd. Ast} = 150 \text{ C/C}$
 $\text{Provide Ast} = 0.0018 \times 1000 \times 344 / 100 = 6.19 \text{ cm}^2$
 $\text{Total Provide } P_t = 0.0022$
 $7.54 \text{ cm}^2 \text{ at Bottom}$
O.K.

ONE-WAY SHEAR CHECK

$V_u = 2.094 \text{ T}$
 $Nominal \text{ Shear Strength}, V_c = \min [(a) , (b) , (c)]$
 $[a] = (0.16 \lambda^* f_c^* 0.5 + 17 P^* V_u^* d_{eff} / (Mu)) * b^* d_{eff}$
 $= (0.16 \times 1 \times 35^* 0.5 + 17 \times 0.0022 \times 2.094 \times 10^4 \times 344 / (6.071 \times 10^7)) \times 1000 \times 344 / 10^4$
 $= 32.715 \text{ Mton}$
 $[b] = (0.16 \lambda^* f_c^* 0.5 + 17 P^*) * b^* d_{eff}$
 $= (0.16 \times 1 \times 35^* 0.5 + 17 \times 0.0022) \times 1000 \times 344 / 10^4$
 $= 33.849 \text{ Mton}$
 $[c] = 0.29 \lambda^* f_c^* 0.5 * b^* d_{eff}$
 $= 0.29 \times 1 \times 35^* 0.5 \times 1000 \times 344 / 10^4$
 $= 59.019 \text{ Mton}$
HENCE, $V_c = 32.715 \text{ Mton}$
 $\Phi^* V_c = 0.75 \times 32.715$
 $= 24.536 \text{ Mton}$
 $\Phi^* (V_c + 0.66 f_c^* 0.5 * b^* d_{eff}) = 0.75 \times (32.715 \times 10^4 + 0.66 \times 35^* 0.5 \times 1000 \times 344) / 10^4$
 $= 103.193 \text{ Mton}$

$\lambda = 1$
 $> 2.094 \text{ Mton}, \text{ Hence O.K.}$
 $> V_u = 2.094 \text{ Mton}, \text{ CROSS SECTION O.K.}$

TWO-WAY SHEAR CHECK

$Area (A) = 4 \text{ m}^2$
 $Perimeter (b_o) = 2 \times ((0.4 + 0.344) + (0.4 + 0.344)) = 2.976 \text{ m}$
 $\beta = 1$
 $\alpha_s = 30 \text{ (Edge Column)}$
 $P_{max} = 16.596 \text{ Mton/m}^2$
 $V_u = ((4 - (0.4 \times 0.4)) \times 16.596) / 63.729$
 $= 63.729 \text{ Mton}$

$Nominal \text{ Shear Strength}, V_c = \min [(a) , (b) , (c)]$
 $[a] = 0.33 \lambda^* f_c^* 0.5 * b_o^* d_{eff}$
 $= 0.33 \times 1 \times 35^* 0.5 \times 2.976 \times 10^3 \times 344 / 10^4$
 $= 199.866 \text{ Mton}$
 $[b] = 0.17 \times (1 + 2/\beta) \lambda^* f_c^* 0.5 * b_o^* d_{eff}$
 $= 0.17 \times (1 + 2/1) \times 1 \times 35^* 0.5 \times 2.976 \times 10^3 \times 344 / 10^4$
 $= 308.884 \text{ Mton}$
 $[c] = 0.083 \times (2 + \alpha_s^* d_{eff} / b_o) \lambda^* f_c^* 0.5 * b_o^* d_{eff}$
 $= 0.083 \times (2 + 30 \times 344 / (2.976 \times 10^3)) \times 1 \times 35^* 0.5 \times (2.976 \times 10^3) \times 344 / 10^4$
 $= 274.86 \text{ Mton}$
HENCE, $V_c = 199.866 \text{ Mton}$
 $\Phi^* V_n = \Phi^* V_c = 0.75 \times 199.866$
 $= 149.9 \text{ Mton}$
 $> 63.729 \text{ Mton}, \text{ Hence O.K.}$

 MA'ADEN معدن شركة التعدين العربية السعودية Saudi Arabian Mining Company   LARSEN & TOUBRO	DESIGN DOCUMENT Ma'aden Document Number: MD-216-39EJ-EG-CV-CAL-1459 Outotec Document Number: 900707_GAU01_AG36_CED01_39304_OU100703853 L&T Document Number: O19045_450_C_K_24434_001	PAGE 11 of 31																																																
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ATTACHMENT 1 : STAAD INPUT FILE OF CIVIL BUILDING																																																		

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 20-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
1 0 0 0; 2 0 5 0; 3 5 5 0; 4 5 0 0; 7 0 0 5; 9 0 5 5; 10 5 5 5; 12 5 0 5;
13 0 -1.5 0; 14 5 -1.5 0; 15 0 -1.5 5; 16 5 -1.5 5;
MEMBER INCIDENCES
1 1 2; 2 2 3; 3 3 4; 4 1 4; 8 7 9; 9 9 10; 10 10 12; 11 7 12; 15 2 9; 16 3 10;
19 1 7; 20 4 12; 21 13 1; 22 14 4; 23 15 7; 24 16 12;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40262
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 3 8 10 21 TO 24 PRIS YD 0.4 ZD 0.4
4 11 19 20 PRIS YD 0.6 ZD 0.25
2 9 15 16 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
13 TO 16 FIXED
DEFINE IBC 2003
SDS 0.1 SD1 0.02 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.18 PZ 0.18
SELFWEIGHT 1
*JOINT WEIGHT
MEMBER WEIGHT
4 11 19 20 UNI 2
FLOOR WEIGHT
*DEAD LOAD
YRANGE 5 5 FLOAD 0.5625
*LIVE LOAD
YRANGE 5 5 FLOAD 0.075
*****
LOAD 1 LOADTYPE Seismic TITLE SLST(+Z)
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic TITLE SLST(+X)
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead TITLE DEAD LOAD
FLOOR LOAD
YRANGE 5 5 FLOAD -0.5625 GY
MEMBER LOAD
4 11 19 20 UNI GY -2
SELFWEIGHT Y -1
LOAD 4 LOADTYPE Live TITLE LIVE LOAD
FLOOR LOAD
YRANGE 5 5 FLOAD -0.15 GY
LOAD 5 LOADTYPE Wind TITLE WIND (+) Z
MEMBER LOAD
3 UNI GZ 0.27
1 UNI GZ 0.27
LOAD 6 LOADTYPE Wind TITLE WIND (+) X
MEMBER LOAD
8 UNI GX 0.27
1 UNI GX 0.27
LOAD 7 LOADTYPE Temperature TITLE THERMAL LOAD
TEMPERATURE LOAD
2 4 9 11 15 16 19 TO 24 TEMP 25
LOAD 8 LOADTYPE Temperature TITLE THERMAL LOAD
TEMPERATURE LOAD
2 4 9 11 15 16 19 TO 24 TEMP -25
LOAD COMB 9 GENERATED LOAD_STRENGTH 1
3 1.4
LOAD COMB 10 GENERATED LOAD_STRENGTH 2
3 1.4 4 1.7
LOAD COMB 11 GENERATED LOAD_STRENGTH 3
3 1.2 4 1.0 5 1.6
LOAD COMB 12 GENERATED LOAD_STRENGTH 4
3 1.2 4 1.0 6 1.6
LOAD COMB 13 GENERATED LOAD_STRENGTH 5
3 1.2 4 1.0 1 1.0
LOAD COMB 14 GENERATED LOAD_STRENGTH 6

```

```

3 1.2 4 1.0 2 1.0
LOAD COMB 15 GENERATED LOAD_STRENGTH 7
3 0.9 5 1.6
LOAD COMB 16 GENERATED LOAD_STRENGTH 8
3 0.9 6 1.6
LOAD COMB 17 GENERATED LOAD_STRENGTH 9
3 0.9 1 1.0
LOAD COMB 18 GENERATED LOAD_STRENGTH 10
3 0.9 2 1.0
LOAD COMB 19 GENERATED LOAD_STRENGTH 11
3 1.2 4 1.6 7 1.2
LOAD COMB 20 GENERATED LOAD_STRENGTH 12
3 1.2 4 1.6 8 1.2
LOAD COMB 21 GENERATED SERVICIBILITY 1
3 1.0
LOAD COMB 22 GENERATED SERVICIBILITY 2
3 1.0 4 1.0 7 1.0
LOAD COMB 23 GENERATED SERVICIBILITY 3
3 1.0 4 1.0 8 1.0
LOAD COMB 24 GENERATED SERVICIBILITY 4
3 1.0 4 0.75 5 0.75
LOAD COMB 25 GENERATED SERVICIBILITY 5
3 1.0 4 0.75 6 0.75
LOAD COMB 26 GENERATED SERVICIBILITY 6
3 1.0 4 0.75 1 0.7
LOAD COMB 27 GENERATED SERVICIBILITY 7
3 1.0 4 0.75 2 0.7
LOAD COMB 28 GENERATED SERVICIBILITY 8
3 0.6 5 1.0
LOAD COMB 29 GENERATED SERVICIBILITY 9
3 0.6 6 1.0
LOAD COMB 30 GENERATED SERVICIBILITY 10
3 0.6 1 0.7
LOAD COMB 31 GENERATED SERVICIBILITY 11
3 0.6 2 0.7
PERFORM ANALYSIS
load list 9 to 20
START CONCRETE DESIGN
CODE ACI
FC 3500 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 MEMB 2 4 9 11 15 16 19 TO 24
MINSEC 10 ALL
SFACE 0.5 MEMB 2 4 9 11 15 16 19 20
EFACE 0.5 MEMB 2 4 9 11 15 16 19 20
DESIGN BEAM 2 4 9 11 15 16 19 20
DESIGN COLUMN 1 3 8 10 21 TO 24
END CONCRETE DESIGN
FINISH

```

 معدن شركة التعدين العربية السعودية Saudi Arabian Mining Company			DESIGN DOCUMENT Ma'aden Document Number: MD-216-39EJ-EG-CV-CAL-1459 Outotec Document Number: 900707_ GAU01_ AG36_ CED01_ 39304_ OU100703853 L&T Document Number: O19045_ 450_ C_ K_ 24434_ 001	PAGE 14 of 31			
Global Project ID Delivery Type Project Name Customer Project Country	900707 ENGINEERING PROCUREMENT CONSTRUCTION MANSOURAH MASSARAH GOLD PROJECT MA'ADEN GOLD AND BASE METALS (MA'ADEN) KINGDOM OF SAUDI ARABIA						
Ma'aden Revision No: Date Revision Text Prepared By Checked Approved	A01 20-10-2020 FIRST ISSUE TEC AVG SJM	C01 20-11-2020 ISSUED FOR CONSTRUCTION TEC AVG SJM	D01 16-10-2023 AS BUILT TEC AVG SJM	Outotec / L&T Revision No.: Date Revision Text Prepared By Checked Approved	0 20-10-2020 FIRST ISSUE TEC AVG SJM	1 20-11-2020 ISSUED FOR CONSTRUCTION TEC AVG SJM	2 16-10-2023 AS BUILT TEC AVG SJM

ATTACHMENT 2 : STAAD OUTPUT FILE OF CIVIL BUILDING

```

*****
*
*          STAAD.Pro V8i SELECTseries5          *
*          Version  20.07.10.64                  *
*          Proprietary Program of                 *
*          Bentley Systems, Inc.                  *
*          Date=    OCT 19, 2020                  *
*          Time=    10:35:24                      *
*
*          USER ID: LNT ECC DIV                   *
*****

```

```

1. STAAD SPACE
INPUT FILE: 5000x5000x5000 Concrete Shed.STD
2. START JOB INFORMATION
3. ENGINEER DATE 20-OCT-20
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER MTON
7. JOINT COORDINATES
8. 1 0 0 0; 2 0 5 0; 3 5 5 0; 4 5 0 0; 7 0 0 5; 9 0 5 5; 10 5 5 5; 12 5 0 5
9. 13 0 -1.5 0; 14 5 -1.5 0; 15 0 -1.5 5; 16 5 -1.5 5
10. MEMBER INCIDENCES
11. 1 1 2; 2 2 3; 3 3 4; 4 1 4; 8 7 9; 9 9 10; 10 10 12; 11 7 12; 15 2 9; 16 3 10
12. 19 1 7; 20 4 12; 21 13 1; 22 14 4; 23 15 7; 24 16 12
13. DEFINE MATERIAL START
14. ISOTROPIC CONCRETE
15. E 2.21467E+006
16. POISSON 0.17
17. DENSITY 2.40262
18. ALPHA 1E-005
19. DAMP 0.05
20. TYPE CONCRETE
21. STRENGTH FCU 2812.28
22. END DEFINE MATERIAL
23. MEMBER PROPERTY AMERICAN
24. 1 3 8 10 21 TO 24 PRIS YD 0.4 ZD 0.4
25. 4 11 19 20 PRIS YD 0.6 ZD 0.25
26. 2 9 15 16 PRIS YD 0.6 ZD 0.3
27. CONSTANTS
28. MATERIAL CONCRETE ALL
29. SUPPORTS
30. 13 TO 16 FIXED
31. DEFINE IBC 2003
32. SDS 0.1 SD1 0.02 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.18 PZ 0.18
33. SELFWEIGHT 1
34. *JOINT WEIGHT
35. MEMBER WEIGHT
36. 4 11 19 20 UNI 2
37. FLOOR WEIGHT

```

****NOTE** about Floor/OneWay Loads/Weights.**

Please note that depending on the shape of the floor you may have to break up the FLOOR/ONEWAY LOAD into multiple commands. For details please refer to Technical Reference Manual Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

```

38. *DEAD LOAD
39. YRANGE 5 5 FLOAD 0.5625
40. *LIVE LOAD
41. YRANGE 5 5 FLOAD 0.075
42. *****
43. LOAD 1 LOADTYPE SEISMIC TITLE SLST(+Z)
44. IBC LOAD Z 1
45. PERFORM ANALYSIS

```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	12	NUMBER OF MEMBERS	16
NUMBER OF PLATES	0	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```

ORIGINAL/FINAL BAND-WIDTH=      8/      5/      30 DOF
TOTAL      PRIMARY LOAD CASES =      1, TOTAL DEGREES OF FREEDOM =      48
TOTAL LOAD COMBINATION CASES =      0 SO FAR.
SIZE OF STIFFNESS MATRIX =      2 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE =      12.0/ 87974.7 MB

```

```

*****
*   IBC 2003 SEISMIC LOAD ALONG Z   :   *
*   CT = 0.016 Cu = 1.700           *
*   TIME PERIODS :                   *
*   Ta = 0.251 T = 0.180 Tuser = 0.180 *
*   TIME PERIOD USED (T) = 0.180     *
*   LOAD FACTOR      = 1.000         *
*   DESIGN BASE SHEAR = 1.000 X 0.040 X 81.79 *
*   = 3.27 MTON                      *
*****

```


46. CHANGE
47. LOAD 2 LOADTYPE SEISMIC TITLE SLST(+X)
48. IBC LOAD X 1
49. PERFORM ANALYSIS

```
*****
*   IBC 2003 SEISMIC LOAD ALONG X   :   *
*   CT = 0.016 Cu = 1.700           *
*   TIME PERIODS :                   *
*   Ta = 0.251 T = 0.180 Tuser = 0.180 *
*   TIME PERIOD USED (T) = 0.180     *
*   LOAD FACTOR      = 1.000         *
*   DESIGN BASE SHEAR = 1.000 X 0.040 X 81.79 *
*                               = 3.27 MTON *
*****
```

50. CHANGE
51. LOAD 3 LOADTYPE DEAD TITLE DEAD LOAD
52. FLOOR LOAD
53. YRANGE 5 5 FLOAD -0.5625 GY
54. MEMBER LOAD
55. 4 11 19 20 UNI GY -2
56. SELFWEIGHT Y -1
57. LOAD 4 LOADTYPE LIVE TITLE LIVE LOAD
58. FLOOR LOAD
59. YRANGE 5 5 FLOAD -0.15 GY
60. LOAD 5 LOADTYPE WIND TITLE WIND (+) Z
61. MEMBER LOAD
62. 3 UNI GZ 0.27
63. 1 UNI GZ 0.27
64. LOAD 6 LOADTYPE WIND TITLE WIND (+) X
65. MEMBER LOAD
66. 8 UNI GX 0.27
67. 1 UNI GX 0.27
68. LOAD 7 LOADTYPE TEMPERATURE TITLE THERMAL LOAD
69. TEMPERATURE LOAD
70. 2 4 9 11 15 16 19 TO 24 TEMP 25
71. LOAD 8 LOADTYPE TEMPERATURE TITLE THERMAL LOAD
72. TEMPERATURE LOAD

73. 2 4 9 11 15 16 19 TO 24 TEMP -25
74. LOAD COMB 9 GENERATED LOAD_STRENGTH 1
75. 3 1.4
76. LOAD COMB 10 GENERATED LOAD_STRENGTH 2
77. 3 1.4 4 1.7
78. LOAD COMB 11 GENERATED LOAD_STRENGTH 3
79. 3 1.2 4 1.0 5 1.6
80. LOAD COMB 12 GENERATED LOAD_STRENGTH 4
81. 3 1.2 4 1.0 6 1.6
82. LOAD COMB 13 GENERATED LOAD_STRENGTH 5
83. 3 1.2 4 1.0 1 1.0
84. LOAD COMB 14 GENERATED LOAD_STRENGTH 6
85. 3 1.2 4 1.0 2 1.0
86. LOAD COMB 15 GENERATED LOAD_STRENGTH 7
87. 3 0.9 5 1.6
88. LOAD COMB 16 GENERATED LOAD_STRENGTH 8
89. 3 0.9 6 1.6
90. LOAD COMB 17 GENERATED LOAD_STRENGTH 9
91. 3 0.9 1 1.0
92. LOAD COMB 18 GENERATED LOAD_STRENGTH 10
93. 3 0.9 2 1.0
94. LOAD COMB 19 GENERATED LOAD_STRENGTH 11
95. 3 1.2 4 1.6 7 1.2
96. LOAD COMB 20 GENERATED LOAD_STRENGTH 12
97. 3 1.2 4 1.6 8 1.2
98. LOAD COMB 21 GENERATED SERVICIBILITY 1
99. 3 1.0
100. LOAD COMB 22 GENERATED SERVICIBILITY 2
101. 3 1.0 4 1.0 7 1.0
102. LOAD COMB 23 GENERATED SERVICIBILITY 3
103. 3 1.0 4 1.0 8 1.0
104. LOAD COMB 24 GENERATED SERVICIBILITY 4
105. 3 1.0 4 0.75 5 0.75
106. LOAD COMB 25 GENERATED SERVICIBILITY 5
107. 3 1.0 4 0.75 6 0.75
108. LOAD COMB 26 GENERATED SERVICIBILITY 6
109. 3 1.0 4 0.75 1 0.7
110. LOAD COMB 27 GENERATED SERVICIBILITY 7
111. 3 1.0 4 0.75 2 0.7
112. LOAD COMB 28 GENERATED SERVICIBILITY 8
113. 3 0.6 5 1.0
114. LOAD COMB 29 GENERATED SERVICIBILITY 9
115. 3 0.6 6 1.0
116. LOAD COMB 30 GENERATED SERVICIBILITY 10
117. 3 0.6 1 0.7
118. LOAD COMB 31 GENERATED SERVICIBILITY 11
119. 3 0.6 2 0.7
120. PERFORM ANALYSIS

121. LOAD LIST 9 TO 20
122. START CONCRETE DESIGN

123. CODE ACI
124. FC 3500 ALL
125. CLB 0.04 ALL
126. CLS 0.04 ALL
127. CLT 0.04 MEMB 2 4 9 11 15 16 19 TO 24
128. MINSEC 10 ALL
129. SFACE 0.5 MEMB 2 4 9 11 15 16 19 20
130. EFACE 0.5 MEMB 2 4 9 11 15 16 19 20
131. DESIGN BEAM 2 4 9 11 15 16 19 20

=====

BEAM NO. 2 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA END	
1	56.	3 - 12MM	0.	4456.	YES	NO
2	544.	2 - 12MM	0.	815.	YES	NO
3	544.	2 - 12MM	3560.	5000.	NO	YES

B E A M N O. 2 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT - Vu= 28.10 KNS Vc= 156.98 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.10 KNS Vc= 156.98 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.

2J	5000X 300X 600	3J
=====		
2No12 H 544.	0.T0 815	2No12 H 544.3560.T0 5000
3No12 H 56.	0.T0 4456	
=====		

00					00	00
2#12					2#12	2#12
3#12	3#12	3#12	3#12	3#12	3#12	
000	000	000	000	000	000	

=====

BEAM NO. 4 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA END	
1	58.	2 - 16MM	0.	4546.	YES	NO
2	544.	4 - 12MM	0.	1231.	YES	NO
3	544.	4 - 12MM	3560.	5000.	NO	YES

B E A M N O. 4 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.

1J	5000X 250X 600	4J
=====		
4No12 H 544. 0.T0 1231	4No12 H 544.3560.T0 5000	
2No16 H 58. 0.T0 4546		
=====		

0000	0000				0000	0000
4#12	4#12				4#12	4#12
2#16	2#16	2#16	2#16	2#16	2#16	
00	00	00	00	00	00	

=====

BEAM NO. 11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR	
					STA	END
1	58.	2 - 16MM	0.	4546.	YES	NO
2	544.	4 - 12MM	0.	1231.	YES	NO
3	544.	4 - 12MM	3560.	5000.	NO	YES

B E A M N O. 11 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.

7J	5000X 250X 600	12J
=====		
4No12 H 544. 0.T0 1231	4No12 H 544.3560.T0 5000	
2No16 H 58. 0.T0 4546		
=====		

0000	0000				0000	0000
4#12	4#12				4#12	4#12
2#16	2#16	2#16	2#16	2#16	2#16	
00	00	00	00	00	00	

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR	
					STA	END
1	56.	3 - 12MM	0.	4456.	YES	NO
2	544.	2 - 12MM	0.	815.	YES	NO
3	544.	2 - 12MM	3560.	5000.	NO	YES

B E A M N O. 15 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT -	Vu=	28.10 KNS	Vc=	156.98 KNS	Vs=	0.00 KNS	
Tu=	0.00 KN-MET	Tc=	6.6 KN-MET	Ts=	0.0 KN-MET	LOAD	9
STIRRUPS ARE NOT REQUIRED.							
AT END SUPPORT -	Vu=	28.10 KNS	Vc=	156.98 KNS	Vs=	0.00 KNS	
Tu=	0.00 KN-MET	Tc=	6.6 KN-MET	Ts=	0.0 KN-MET	LOAD	9
STIRRUPS ARE NOT REQUIRED.							

2J	5000X	300X	600	9J
2No12 H 544.	0.TO	815		
			2No12 H 544.3560	.TO 5000
3No12 H 56.	0.TO	4456		

00 2#12						00 2#12	00 2#12
3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR	
					STA	END
1	56.	3 - 12MM	0.	4456.	YES	NO
2	544.	2 - 12MM	0.	815.	YES	NO
3	544.	2 - 12MM	3560.	5000.	NO	YES

AT START	SUPPORT -	Vu=	28.10 KNS	Vc=	156.98 KNS	Vs=	0.00 KNS	
Tu=	0.00 KN-MET	Tc=	6.6 KN-MET	Ts=	0.0 KN-MET	LOAD	9	
STIRRUPS ARE NOT REQUIRED.								
AT END	SUPPORT -	Vu=	28.10 KNS	Vc=	156.98 KNS	Vs=	0.00 KNS	
Tu=	0.00 KN-MET	Tc=	6.6 KN-MET	Ts=	0.0 KN-MET	LOAD	9	
STIRRUPS ARE NOT REQUIRED.								

3J	5000X	300X	600	10J
2No12 H 544.	0.T0	815		
3No12 H 56.	0.T0	4456		

00 2#12						00 2#12	00 2#12
3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	3#12 000	

=====

BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR	
					STA	END
1	58.	2 - 16MM	0.	4546.	YES	NO
2	544.	4 - 12MM	0.	1231.	YES	NO
3	544.	4 - 12MM	3560.	5000.	NO	YES

B E A M N O. 19 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.

1J	5000X 250X 600	7J
=====		
4No12 H 544. 0.T0 1231	4No12 H 544.3560.T0 5000	
2No16 H 58. 0.T0 4546		
=====		

0000	0000				0000	0000
4#12	4#12				4#12	4#12
2#16	2#16	2#16	2#16	2#16	2#16	
00	00	00	00	00	00	

=====

BEAM NO. 20 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 34. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA END	
1	58.	2 - 16MM	0.	4546.	YES	NO
2	544.	4 - 12MM	0.	1231.	YES	NO
3	544.	4 - 12MM	3560.	5000.	NO	YES

B E A M N O. 20 D E S I G N R E S U L T S - S H E A R

AT START SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 47.31 KNS Vc= 152.41 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 9
STIRRUPS ARE NOT REQUIRED.

4J	5000X 250X 600	12J
=====		
4No12 H 544. 0.T0 1231	4No12 H 544.3560.T0 5000	
2No16 H 58. 0.T0 4546		
=====		

0000	0000				0000	0000
4#12	4#12				4#12	4#12
2#16	2#16	2#16	2#16	2#16	2#16	
00	00	00	00	00	00	

*****END OF BEAM DESIGN*****

132. DESIGN COLUMN 1 3 8 10 21 TO 24

=====

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

8 - 16 MM	1.005	9	END	0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)				
TIE BAR NUMBER 10 SPACING 256.00 MM				

=====

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

8 - 16 MM	1.005	9	END	0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)				
TIE BAR NUMBER 10 SPACING 256.00 MM				

=====

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	9	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)

TIE BAR NUMBER 10 SPACING 256.00 MM

=====

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	9	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)

TIE BAR NUMBER 10 SPACING 256.00 MM

=====

COLUMN NO. 21 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1817.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	19	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)

TIE BAR NUMBER 10 SPACING 256.00 MM

=====

COLUMN NO. 22 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1817.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	19	STA	0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)				
TIE BAR NUMBER 10 SPACING 256.00 MM				

=====

COLUMN NO. 23 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1817.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	19	STA	0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)				
TIE BAR NUMBER 10 SPACING 256.00 MM				

=====

COLUMN NO. 24 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1817.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	19	STA	0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)				
TIE BAR NUMBER 10 SPACING 256.00 MM				

*****END OF COLUMN DESIGN RESULTS*****

133. END CONCRETE DESIGN
134. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= OCT 19,2020 TIME= 10:35:37 ****

* For technical assistance on STAAD.Pro, please visit *
* <http://selectservices.bentley.com/en-US/> *
* *
* Details about additional assistance from *
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